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B. Choi, *Introduction to Python Network Automation*

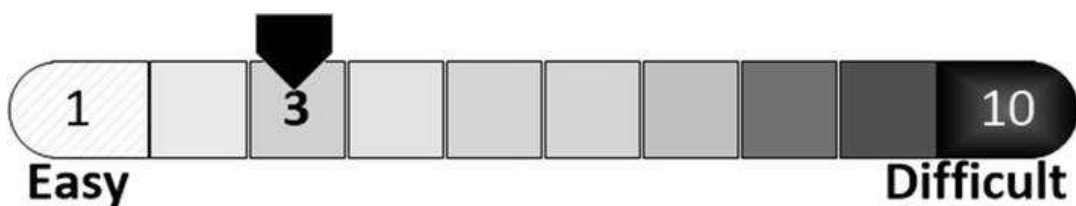
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4. Introduction to VMware Workstation

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VMware Workstation Pro is one of the most popular end-user desktop virtualization solutions used by enterprise IT personnel. It allows IT engineers to build and test real-world-like virtual machines and network devices from their own PC/laptop. Learning to build a fully functioning and integrated virtual lab on a single computer and master Python network automation basics on a single PC is this book's underlying goal. This chapter is designed to teach you the basics of the most popular and versatile VMware product, VMware Workstation 15 Pro, and basic virtualization concepts. After reading this chapter, you will gain the following knowledge: you'll understand the difference between Type-1 and Type-2 hypervisors, understand various IT vendors offering different desktop virtualization solutions, learn how to install VMware Workstation, learn how to perform general administration on VMware Workstation, and understand how VMware Workstation's network adapters operate.



VMware Workstation at a Glance

VMware was created in Palo Alto, California, in 1998, and the company has been growing and leading the enterprise virtualization market for more than two decades. With the exceptions of the first three chapters in this book, this book will rely on VMware's desktop virtualization technology, namely, VMware Workstation 15 Pro. Before embarking on your journey into Python automation, you must approach your studies with an open mind into the various IT technologies. Learning to write Python code does not guarantee you will be successful with your network automation attempts. You will need to learn about regular expressions and Linux administration, which are topics often outside of a network engineer's comfort zone but are essential to writing functional code. That's right; it is a rocky road to Python network automation. For a novice Python learner, learning Python syntax on its own may take months, and it may take years to be able to call yourself an expert in that programming language. Still, the more significant challenge is applying Pythonic concepts and syntax to your work. You need to learn and build Python knowledge on top of your current networking and general IT knowledge. Writing Python code is only the first step of your journey. It requires you to better understand Linux, networking, simple mathematics, regular expressions, interaction with several APIs, and much more.

With limited time and resources, you are trying to learn essential skills required to make the right start in network automation using Python, where all the learning will take place on a single Windows PC (or a laptop). To do this, we must leverage the power of VMware's desktop virtualization solution. The reason for using VMware Workstation 15 is straightforward. It ticks all the boxes for novice Python coders like you to make a practical yet straightforward virtual lab for learning in multiple scenarios.

First, we have to get comfortable with the basic virtualization concepts and some basic functions of VMware Workstation 15. Even if you

are a longtime user of various virtualization solutions, including VMware products, there will be something new that you will take away from this chapter, so I hope you can read this chapter in its entirety. After reading this chapter, you will have no problem following the rest of the book's content and eventually building a Python network automation lab. In the previous chapter, you installed Python 3 on your host Windows PC and learned Python basics on Windows 10. However, in this chapter and the coming chapters, you will be guided through installing two flavors of Linux distribution servers on the same Windows host using VMware Workstation 15 Pro. Therefore, you can also comfortably learn the Python basics of Linux servers and start picking up Linux basic administration skill sets in Chapters [7](#) and [8](#). To run various operating systems, we have to use virtualization technology to help us run multiple machines on the same host platform; VMware Workstation 15 can help us. As part of the pre-tasks for this chapter, you will download and install VMware Workstation 15 on your PC. After that, you will learn the basics of VMware Workstation 15 and then install the latest version of the Ubuntu 20 and CentOS 8 servers in the following two chapters. Throughout the book, these servers will be used to teach you Linux administration, Python installation, and Python module installation, and how to build the CentOS server as an all-in-one IP services server, which can be used as a pocket knife tool in our proof-of-concept (PoC) labs at the end of this book.

The more you know about VMware Workstation 15's features and basic virtualization functions, the more you will become more effective in building this testing lab environment. When designing and building a PoC lab for enterprise IT solutions, often expensive and extremely powerful devices are deployed only to test the most straightforward concept. Sometimes real equipment is required and justified, for example, for stress testing in hardware-related system performance tests. Also, the procurement process to purchase the right equipment can take weeks, if not months. And what about finding the rack space with the sufficient power voltage and cooling to keep your lab running

smoothly? Also, physical labs involve the right cables with the correct connector types to suit the vendor's equipment interfaces, which can slow you down while building your company's PoC lab. Too often, thousands of dollars are wasted on a company lab when it is only utilized by a couple of people in your company. This is a total waste of the company's money, and definitely, there is no green IT on this side of the court. Thanks to advancements in virtualization technologies in the past decade, more and more equipment is moving to virtualization platforms, as you have seen with the explosion of cloud-based computing and software-as-a-service IT offerings, and anything IT-as-a-service leverages off the power of virtualization technology in one form or another.

VMware Workstation 15 Pro is VMware's most popular, cross-platform desktop virtualization program that supports both Windows and Linux. For the macOS operating system, VMware Fusion provides the same features. There is a VMware Workstation Player for the free version, but it lacks all the features you need to make your lab a complete lab. For this reason, you will use VMware Workstation 15 Pro. You will install a series of software applications, one at a time, on the Microsoft Windows 10 host system. According to Wikipedia, the 2017 figures suggested that the percentage of desktop/laptop users using the Windows operating system accounts is around 88 percent, macOS users account for roughly 10 percent, and finally, there are only 2 percent of Linux desktop/laptop users in the world. From my own experience, even within the IT industry, similar user percentages are reflected. However, with enterprise-level automation solutions based on Python as the preferred programming language, Linux is the default operating system for added flexibility, higher performance, low cost of ownership, and better security.

You can download Workstation 15 Pro Evaluation for Windows from VMware's official website and take advantage of the 30-day free trial period. After 30 days, you can purchase a 15.5 version license and permanently activate the evaluation software. When writing this book, the latest VMware Workstation for Windows version was 15.5.1, build 15018445, but your

downloaded version will be newer (version 15.5 or even version 16.x), but it should still work with most of the software recommended in this book. For your reference, version 13 of VMware Workstation has never been released, and version 14 was released, but it had multiple bugs with GNS3 integration. Strangely enough, GNS3 and Cisco VIRL IOS software will work best on either VMware Workstation 12.5.x or 15.x, but not with the 14.x version. So if you are using VMware 14, you must upgrade to the latest VMware Workstation 15.x version to avoid any problems. It is also recommended that you disable auto-update features to keep your version of software stable at all times. This compatibility recommendation is based on my software compatibility testing. Figure 4-1 shows the product information for VMware Workstation 15 Pro used in this book (filename: `VMware-workstation-full-15.5.1-15018445.exe`).



Figure 4-1. About VMware Workstation 15 Pro

Even if you are not yet familiar with the concept of virtualization and this software, you will not have any problem following this chapter's content, as this book assumes that you have had no previous experience with this VMware product. Those who are not familiar with a desktop virtualization program such as this still can read through the chapter to get familiar with this software and some basic desktop virtualization concepts. For those who have many years of virtualization

experience, please skim through this chapter and jump to the next two chapters, which cover installing Ubuntu 20.04 and CentOS 8 Linux virtual machines (VMs).

Type-1 vs. Type-2 Hypervisors

The first thing to understand about virtualization (hypervisor) software is the difference between Type-1 and Type-2 hypervisors. To explain the difference between Type-1 and Type-2 virtualization software, if the hosting hardware does not require separate operating system software, the virtualization software is executed directly on the hosting hardware. It is classified as a Type-1 hypervisor. If the hosting hardware requires separate operating system software, it is classified as a Type-2 hypervisor. To put it more bluntly, the Type-1 hypervisor is typically used for business use, and the Type-2 hypervisor is designed and used for personal and experimental use. That doesn't mean that Type-2 shouldn't be used for business use. However, there is very little reason to use Type-2 hypervisors over Type-1 in a commercial environment as Type-1 hypervisors provide better stability, high availability, and better scalability features. The Type-1 hypervisor family, also known as *bare metal*, refers to enterprise-level server platforms released without built-in OS software. Typical Type-1 hypervisor products include VMware's ESXi, Microsoft's Hyper-V for Servers, Citrix's XenServer, and Oracle VM.

In contrast, examples of Type-2 hypervisors being developed for desktop applications include VMware's VMware Workstation, Fusion, Player, Oracle's VM VirtualBox (formerly Sun Microsystems), Microsoft's VirtualPC, and Red Hat's Enterprise Virtualization. Type-2 hypervisors are also known as desktop (or hosted) hypervisors. VMware Workstation 15 Pro, used in this book, runs on top of a Windows 10 host machine, making it a Type-2 hypervisor.

VMware Workstation Pro for a Proof-of-Concept Lab

VMware Workstation 15 Pro is VMware's flagship desktop virtualization program. Version 1 was released in 1999, and the latest version was 16 at the time of authoring this book. Although this book is based on version 15.x, you will not have a problem building the lab based on version 16.x. Installing this desktop program and using it are intuitive tasks, and it can be used without user training. It supports both Windows and Linux operating systems. Virtual programs running on top of the operating system, such as VMware Workstation, are sometimes called *experimental virtualization programs*, and the latest software trial versions are available from VMware's official download page. The basic installation is simple, and the installation gets completed by pressing a few buttons. If you do not have a valid license, the program will run on a 30-day trial license. After a 30-day trial period, you can continue using the software after purchasing a license.

Of course, while choosing desktop virtualization software for lab use, you will find other popular software from Microsoft and Oracle. Oracle's VirtualBox and Microsoft's Hyper-V were considered a replacement of VMware Workstation, but because of software compatibility and system stability, we will use VMware Workstation 15 Pro. Enterprise-level virtualization programs such as VMware's ESXi and Microsoft's Hyper-V were considered, but Type-1 hypervisors such as these have higher hardware specification requirements. For example, Type-1 hypervisor software needs to be installed on hypervisor-specific hardware (bare-metal) servers. Also, you will need a separate client PC to remote console to the server to operate the Type-1 hypervisor software. Also, Type-1 hypervisors usually get deployed in data centers or company communication rooms and are not meant to be mobile.

Before Using VMware Workstation

When installing Type-1 hypervisor software such as ESXi 6.5, there are a few stringent hardware requirements during the installation process. The software has a built-in process to check the minimum

hardware requirements. Luckily, for VMware Workstation 15 Pro installation on your host's operating system, there are no or minimal hardware prerequisites. If your computer's CPU and motherboard support virtualization technology, then you can start the installation. The BIOS of the newer PCs or laptops come out of the factory with the Virtualization Technology option enabled. If you are using an older type of PC or laptop, this setting needs to be enabled, but this is the only thing you have to change before installing the software on your PC. Until recently, VMware's virtualization solutions tended to favor Intel's CPU architecture slightly more than AMD's CPU architecture. Intel CPUs are far superior to AMD CPUs in terms of performance, compatibility, and efficiency in my own experience. Of course, AMD has invested heavily in CPU development lately, but AMD CPUs are better geared toward individual users who enjoy PC gaming.

Before installing VMware Workstation 15 Pro, ensure that the virtualization support is set to Enabled after entering the BIOS of your laptop/PC motherboard, as shown in Figure 4-2.

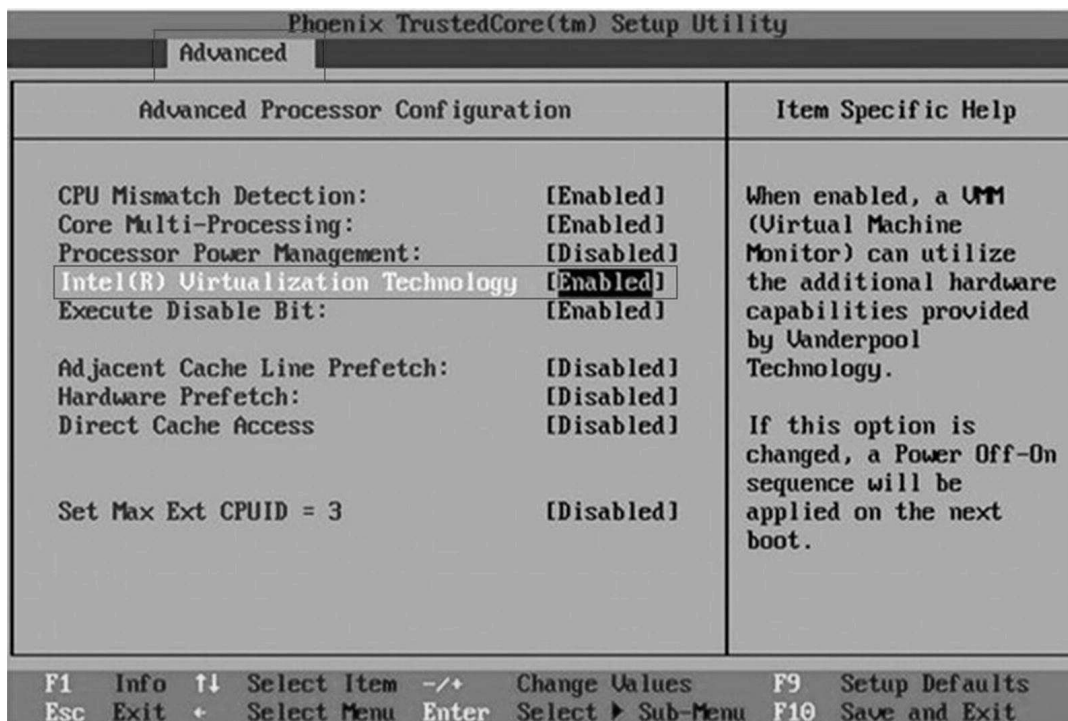


Figure 4-2. Checking the Intel CPU motherboard virtualization support setting

If the use of Intel VT (or AMD-V) is set to Disabled in the BIOS, change the setting to Enabled, and then press the F10 key to save the changes and log into Windows 10 normally. Tables [4-1](#) and [4-2](#) list the BIOS entry keys for the major motherboard and laptop manufacturers. The function key or key combo to enter the BIOS varies from one manufacturer to another, and the most common manufacturers' keys are provided for your reference. If you can't find your laptop manufacturer in the table, go to the manufacturer's website and locate the BIOS information to change the BIOS settings.

Table 4-1. BIOS Setup Entry Key by Laptop Manufacturer

Manufacturer	BIOS Key Combo
Acer	F2/Del (new) F1/Ctrl+Alt+Esc (old)
Asus	F2/F10/Del/Insert/Alt+F10
Compaq	F1/F2/F10/Del
Dell	F1/Del/F12/F3 (new) (press when the Dell logo appears on the screen) Ctrl+Alt+Enter/Fn+Esc or Fn+F1/Ctrl+F11 (old)
eMachines	F2/Del
Fujitsu	F2
Gateway	F1/F2
HP	F1/F2/F6/F9/F10/F11/Esc Tablet PC: F10/F12
Lenovo	F1/F2/F11 (new) Ctrl+Alt+F3/Ctrl+Alt+Ins/Fn+F1 (old)
LG	F10/F11/F12

Manufacturer	BIOS Key Combo
Samsung	F2/F4
Sony	F1/F2/F3/Assist
Toshiba	F2/F1/ESC/F12
Other	Refer to the respective vendor website

Table 4-2. BIOS Setup Entry Key by Motherboard

Manufacturer	BIOS Key/Key Combo
ASROCK	F11
ASUS	F12/F2/F8/F9
BIOSTAR	F7/F9
Compaq	F10
EMTec	F9
Foxconn	F7/Esc
GIGABYTE	F12
HP	F9
Intel	F10
LG	F12
MSI	F11

Manufacturer	BIOS Key/Key Combo
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Pegatron	F11
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Samsung	ESC
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Other	Refer to the respective vendor website
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Here's one more reminder on the VMware Workstation version: the latest version of VMware Workstation 15 Pro at the time of publication was 15.5.1 build-15018445. Also note that version 14 showed compatibility issues with some applications used in this book. The problem was identified as the security file and VIRT integration requirement for the GNS3 Docker program download process. Please use the latest 15.5 version, which is compatible with GNS3. This book uses version 15.5.1, the latest version of 15.5 at the time of writing this chapter. If you do not have a copy of version 15, then use version 12. Version 12 is also compatible and stable with GNS3, VIRT software, and GNS3 Docker images. After the first draft and technical review of this book, VMware released version 16. You should be able to use version 16 or the latest version and still follow the contents of this book. For now, please use VMware Workstation 15.x Pro or 16.x or 12.x but not 14.x.



Download and install VMware Workstation 15 (or 16) Pro on your Windows PC before reading this chapter. The download and installation is straightforward, but in case you need help, the link to download the installation guide is provided with this book.

Download VMware Workstation 15/16 from the following URL:

URL: <https://www.vmware.com/au/products/workstation-pro/workstation-pro-evaluation.html>

The “Chapter 4 Pre-task – Installing VMware Workstation” guide is available from here:

URL: https://github.com/pynetauto/apress_pynetauto

What's Next on VMware Workstation 15 Pro?

As a pre-task to this chapter, you must install VMware Workstation 15 Pro on your host PC. We will quickly review the menu options for those who are new to virtualization technology and this program before installing two Linux virtual servers and importing the GNS3 VM. If you are confident that you know the VMware Workstation feature inside and out, skip to Chapter 5. Anyway, you can think of a virtual machine (VM) as another independent computer running on a host computer's operating system (Type-2). Users can interact with the VM through the Workstation 15 Pro's main user console and use various operating system installation media files to initially create and install VMs and then configure, import, export, control, and manage all aspects of the VMs. You already saw the main user console of VMware Workstation 15 Pro after the pre-task, and you already noticed that the user interface is concise and intuitive. However, the operation and behavior of VMs are much more sophisticated than you might think at first. A VM looks like an application, but it is a computer with many of the native features of a real hardware computer. The best way to get to know Workstation 15 Pro is to use it, but to make your learning a little easier, we will go through some console-related terminology in the next section. Let's look at the user interfaces, menus, and the Virtual Network Editor.

After a quick review, you will be guided to links to download and install a couple of VMs and also download and import a VM using the GNS3 .ova file. You must create all three VMs by the end of this chapter as the rest of the book will rely on these VMs to learn Linux, learn regular expressions, install Python and network-related modules, and learn how to write Python network automation scripts in the virtual labs. To emulate the real lab scenarios, you will test various networking concepts and get familiar with the most commonly used file-sharing services. For instance, this book's last few chapters will have lab

scenarios involving FTP, SFTP, and TFTP servers based on a CentOS 8 virtual machine.

Optionally, if you want to create new VMs here and then export them to a real production environment, you can do so by using the VMware Converter software. VMs can be prestaged on VMware Workstation and then converted later, converting the VMs to Type-1 VMs running on VMware's vSphere 6.5. For now, let's first look at the main user console and its menus of Workstation 15 Pro.

VMware Workstation 15 Pro User Console

The Workstation 15 Pro user window (or console) lets you install, control, and manage various types of VMs. Figure 4-3 shows the main user interface and default buttons of VMware Workstation 15 Pro. This figure will provide you with information on various features of menus and keys. No special skills are required to use this program, and you can easily create and use a VM after installation. Now, let's learn the menus and key features.

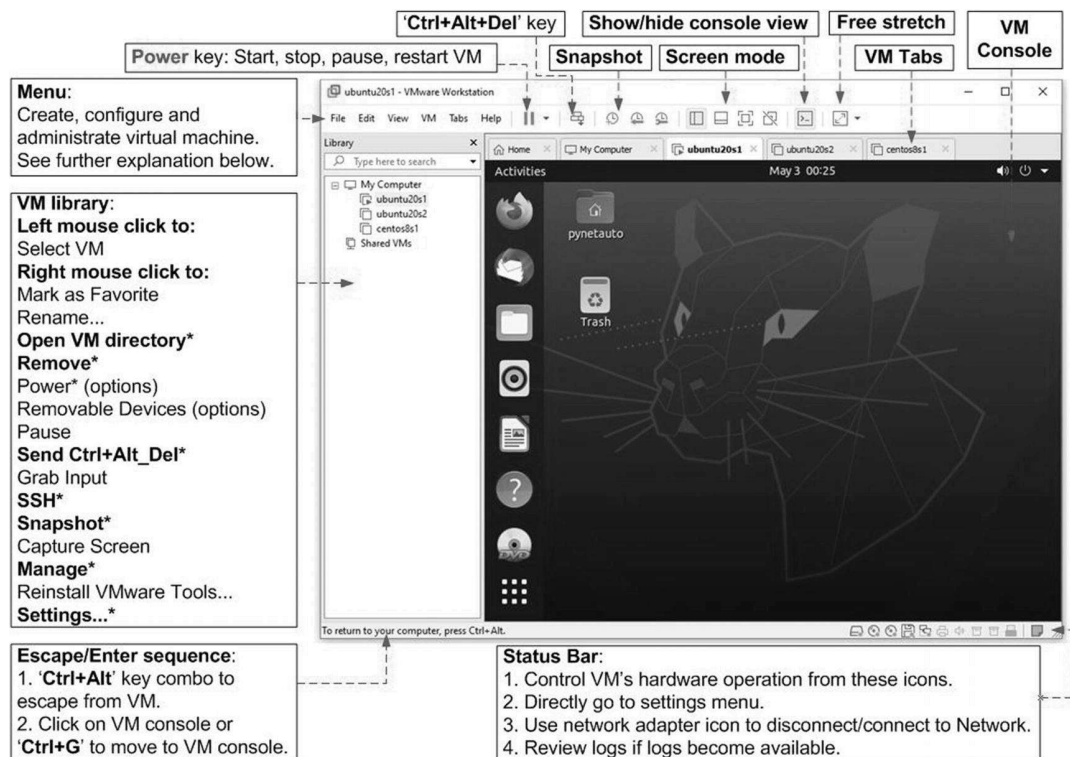


Figure 4-3. VMware Workstation, user window

Basic Operations of VMware Workstation 15 Pro

Figure 4-4 depicts the main user interface for VMware Workstation 15 Pro, and this is the primary interface where you will manage your VMs. Compared to Microsoft Hyper-V on Windows 10 and Oracle VirtualBox, VMware Workstation Pro has the most friendly user interface and powerful features at the user's fingertips. Let's quickly learn the most basic functions of this program.

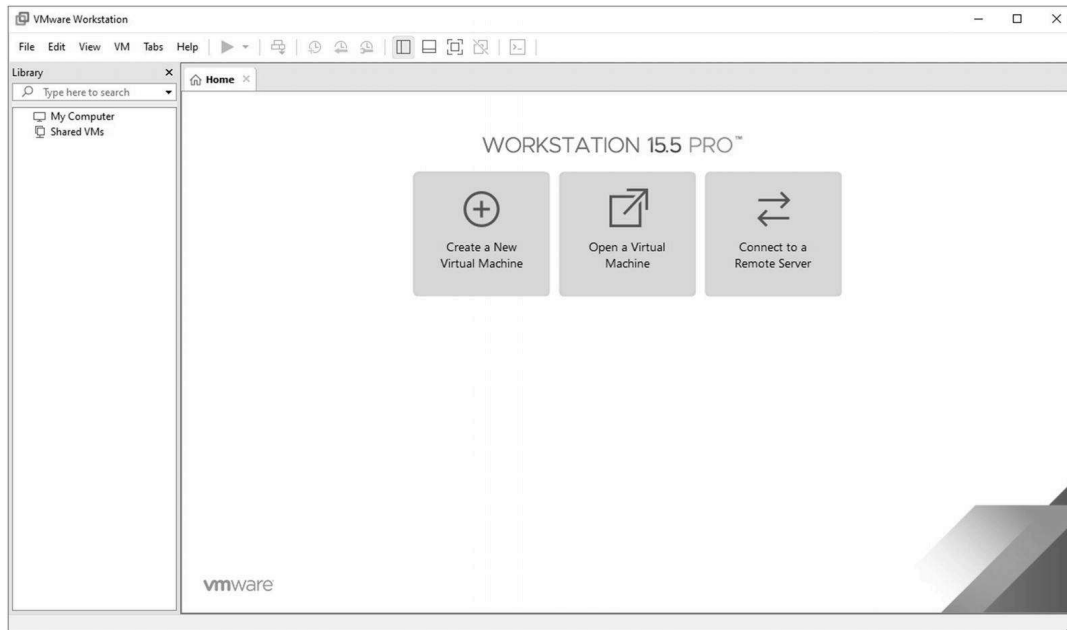


Figure 4-4. VMware Workstation, main user window

VMware Workstation Pro: Basic Operations

To start VMware Workstation 15 Pro (Figure 4-5a):

You can start the program by using the VMware shortcut on the Windows desktop or Windows 10's Start menu: Start ► Programs ► VMware ► VMware Workstation.

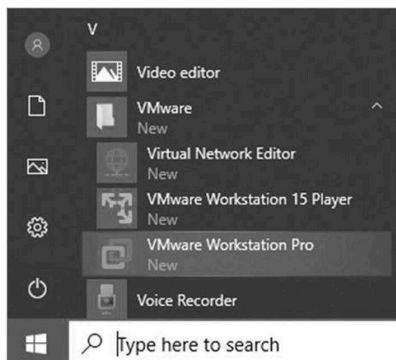
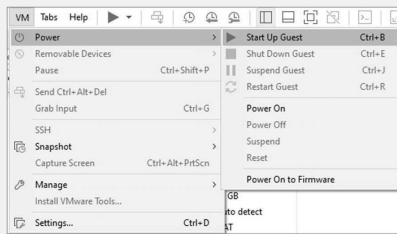


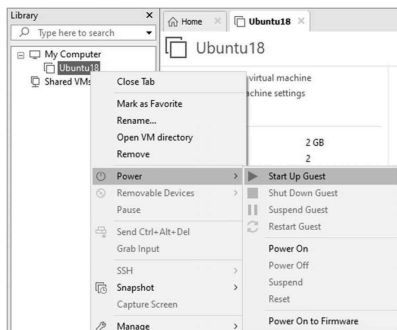
Figure 4-5a. Starting VMware

Workstation Pro

**Figure 4-5b.** Power Options 1

To power on/off, suspend or reset a VM (Figure 4-5b):

One way to power on/off or pause a VM is to select a VM from the VM menu: VM ► Power ► Start Up Guest.

**Figure 4-5c.** Power Options 2

The second way to power on/off, suspend or reset a VM is to select a VM on VM Library (Figure 4-5c).

Right-click a VM and select Power ► Start Up Guest.

**Figure 4-5d.** Power Options 3

The third way to power on/off, suspend or reset a VM is to use the power key options on the top of the user window (Figure 4-5d).

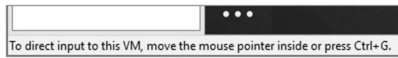


Figure 4-5e. To move to a VM

To move to a VM (from the host PC), press Ctrl+G.

To use a running VM, click the VM console using the left mouse or press Ctrl+G on your keyboard (Figure [4-5e](#)).



Figure 4-5f. To exit a VM

To exit a VM (to move the cursor back to the host PC), press Ctrl+Alt. To exit the mouse cursor back to the Windows host while using the VM, press the Ctrl+Alt keys to exit the host (Figure [4-5f](#)).

For more detailed user instructions, please refer to

<https://docs.vmware.com/en/VMware-Workstation-Pro/15.0/workstation-pro-15-user-guide.pdf>.

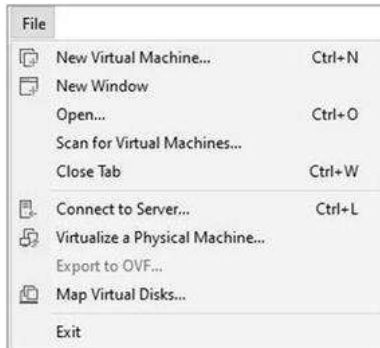
VMware Workstation Menu

Table [4-3](#) shows all the menus at a glance. The most commonly used features are marked with an asterisk (*). You may find it helpful to check the features and skip to creating virtual machines.

Table 4-3. Menus at a Glance

Menu Screen Capture

Menu Function



File menu

File menu:

Create a new VM

Open a new window

Open a VM*

Scan for VMs

Close tab

Connecting to external servers, vCenter, and vSphere servers

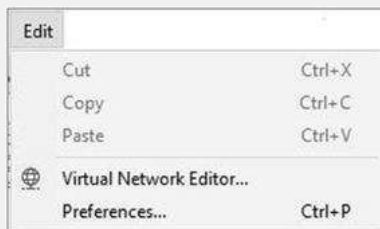
Connect to vCloud Air

Create p2v (physical to virtual)

Export to OVF file*

Connecting to a virtual disk

Exit*



Edit menu

Edit menu:

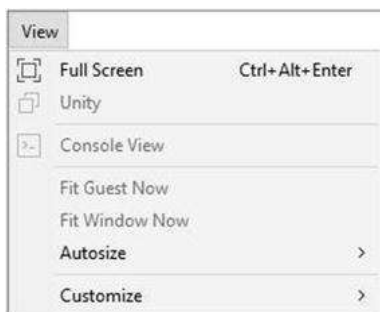
Cut

Copy

Paste

Virtual Network Editor*

Preferences*



View menu

View menu:

View in full-screen*

Unity mode

Console view

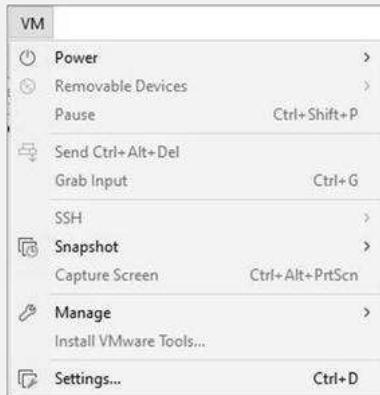
Fit screen to guest OS

Fit screen to window

Auto size

Custom view

Menu Screen Capture



VM menu

Menu Function

VM (Virtual Machine):

Power*

Removable devices

Pause

Send Ctrl+Alt+Del key*

Catch typing

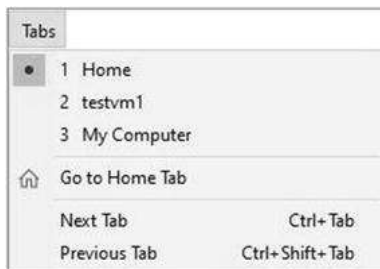
Snapshot Management*

Screen capture

Manage

VMware Tools installation

Settings*



Tabs menu

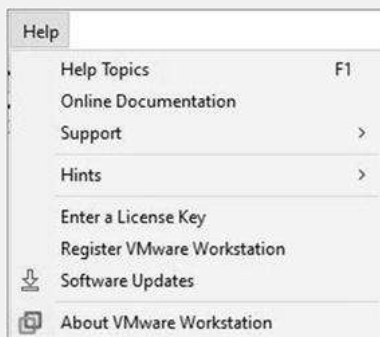
Tabs Menu:

Home

Go to Home Tap

Next tab

Previous tab



Help menu

Help:

Help

Online documentation

support

hint

Enter License Key*

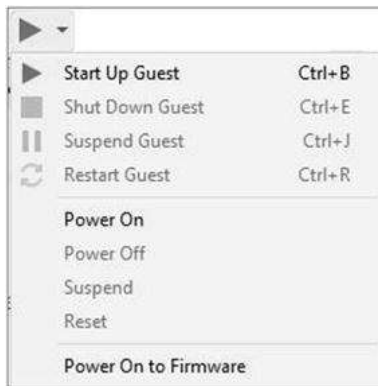
VMware Workstation Registration

Software update

About VMware Workstation

Menu Screen Capture

Menu Function



VM power icon

VM power icon:

Guest On*

Guest Off*

Guest Suspend

Resume Guest*

Power on

Power off

Pause

reset

Enters BIOS when powered on *



Send Ctrl+Alt+Del icon

Send Ctrl+Alt+Delete to the VM



Snapshot icons

Take a snapshot of this VM

Revert this VM to its parent snapshot

Manage snapshots for this VM

Screen mode option
icons

Show or hide the library

Show or hide the thumbnail bar

Enter full-screen mode

Enter Unity mode

Show or hide Console
view icon

Show or hide console view

Menu Screen Capture

Menu Function



Free stretch

Free stretch icon

Virtual Network Adapters

One of the most common questions beginners ask about VMware Workstation 15 Pro is how to use different virtual network adapter settings. It is difficult to provide an answer in a nutshell; you could read the unnecessarily lengthy documents on the VMware website, but most of us do not have hours to read vendor-provided documentation. Here, we will cover the very basics to get started with the VM network. First-time users need to know how each virtual network adapter differs from one another and how to use each one. The best way to explain this is to review which virtual network adapters are available to a user. Try to understand each network adapter types' definition and then look at different adapter settings in your programs to reveal how each adapter is configured on your PC. First, let's look at the Virtual Network Editor to review out-of-the-box virtual network adapter settings.

Virtual Network Editor Overview

VMware Workstation provides virtual network features in virtual switches, where three virtual switches are mapped to three specific networks and up to 17 virtual switches can be created by the user as needed. The different networking types will be discussed in detail in the next section. So, you can have up to 20 virtual switches (VMnet0 through VMnet19), with each virtual switch representing a single flat network or a subnet. Like in a real production environment, you can connect multiple VMs to the same network (virtual switch). First, open

the Virtual Network Editor menu to review which default connection types are available.

Task

1a When starting VMware Workstation Pro, it is a best practice to start the program by right-clicking the desktop icon and selecting the “Run as administrator” option, as shown in Figure 4-6. This way, you are launching the program with a full administrator-level privilege to change any program settings.

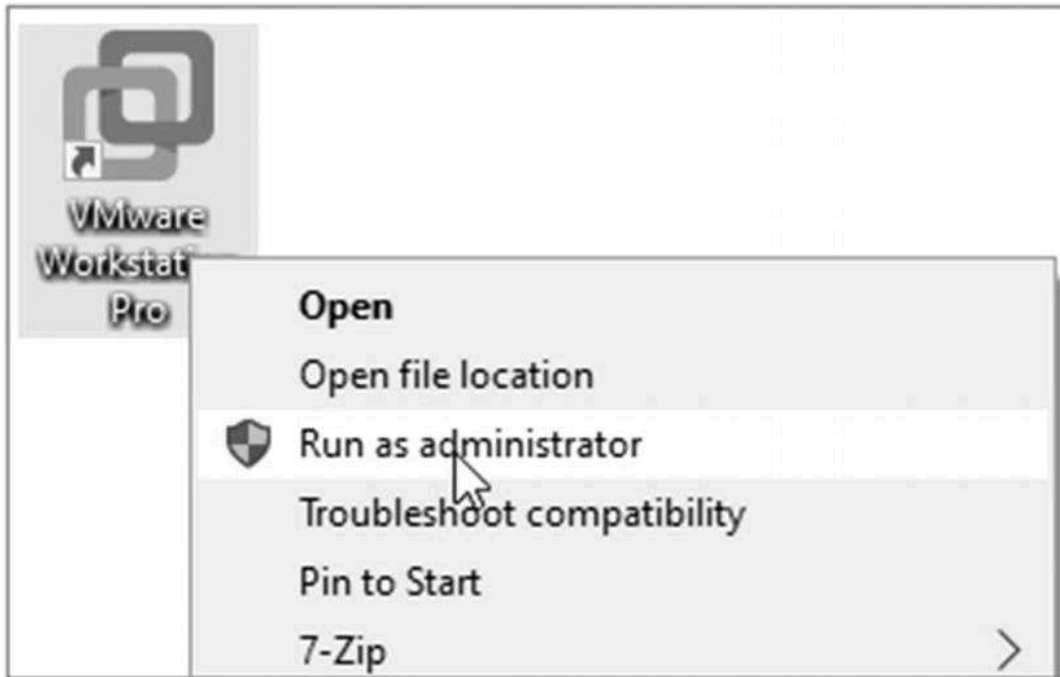


Figure 4-6. VMware Workstation Pro, running as administrator

1b If you don’t want to click the “Run as administrator” option every time you start the program, you can change the shortcut properties to make the program run in administrator mode. Go to VMware Workstation Pro’s Desktop icon and right-click your mouse and select Properties, as shown in Figure 4-7.

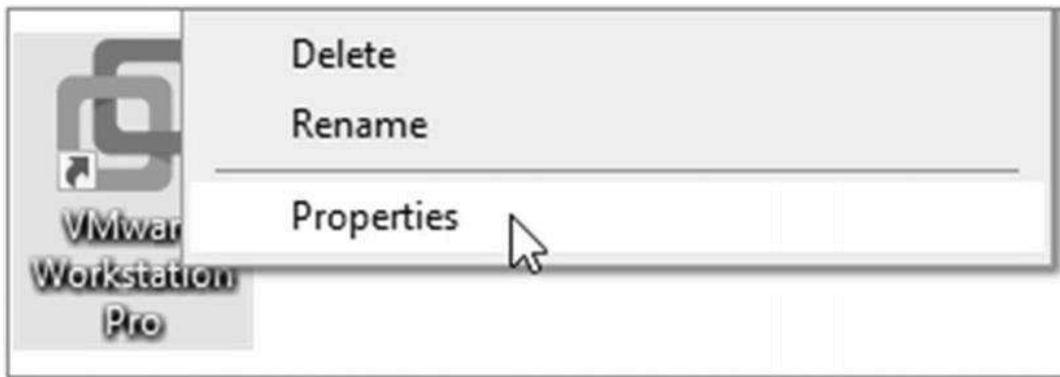


Figure 4-7. VMware Workstation Pro, shortcut icon Properties

In the shortcut properties menu, click Compatibility, and under the Settings option, check “Run this program as an administrator” and then click the Apply and OK buttons. Now your program will always run as an administrator (see Figure [4-8](#)).

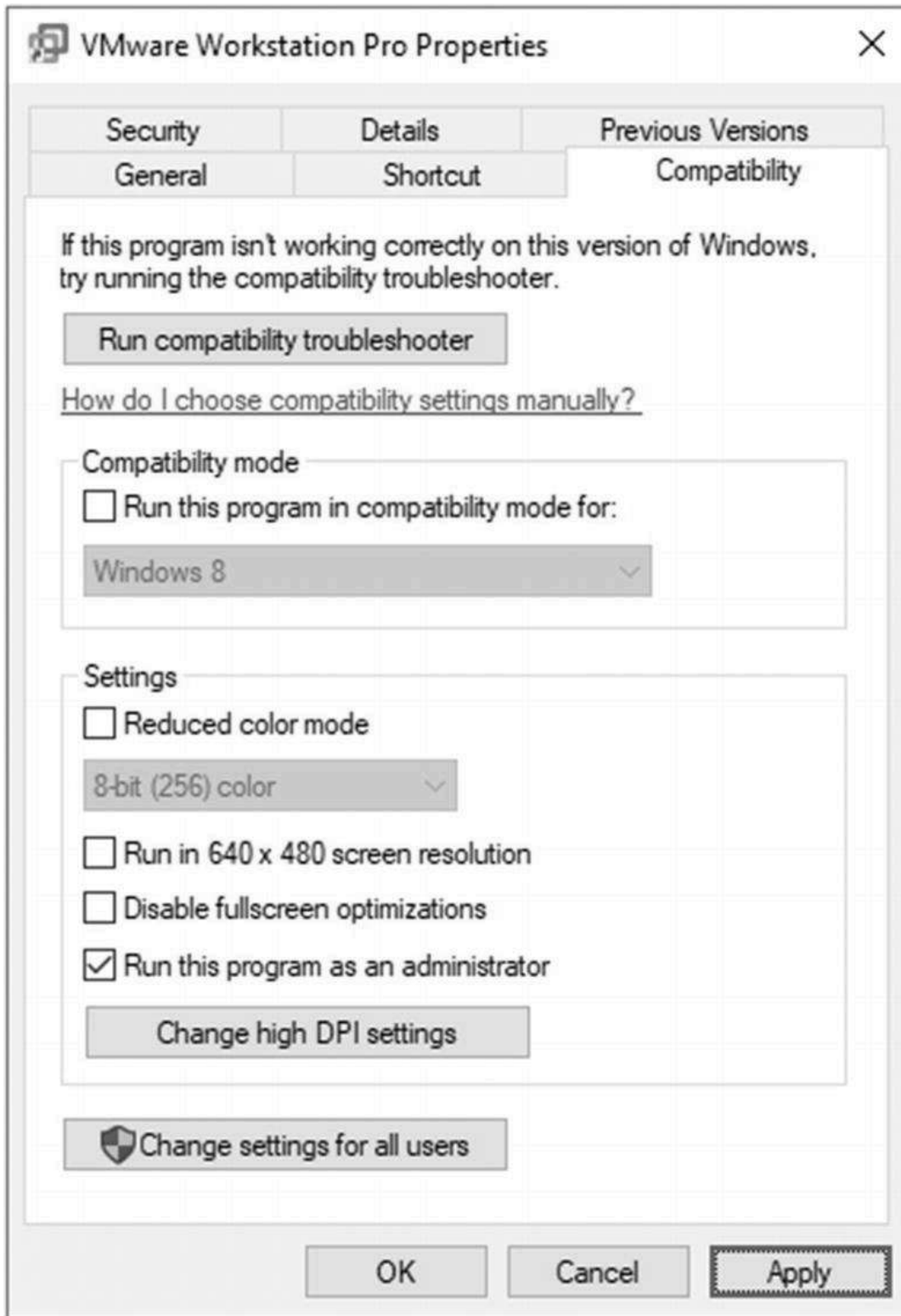


Figure 4-8. VMware Workstation Pro, changing the properties Compatibility settings

2 Now, to open and view virtual network adapters from the menu, select Edit ► Virtual Network Editor (Figure [4-9](#)).

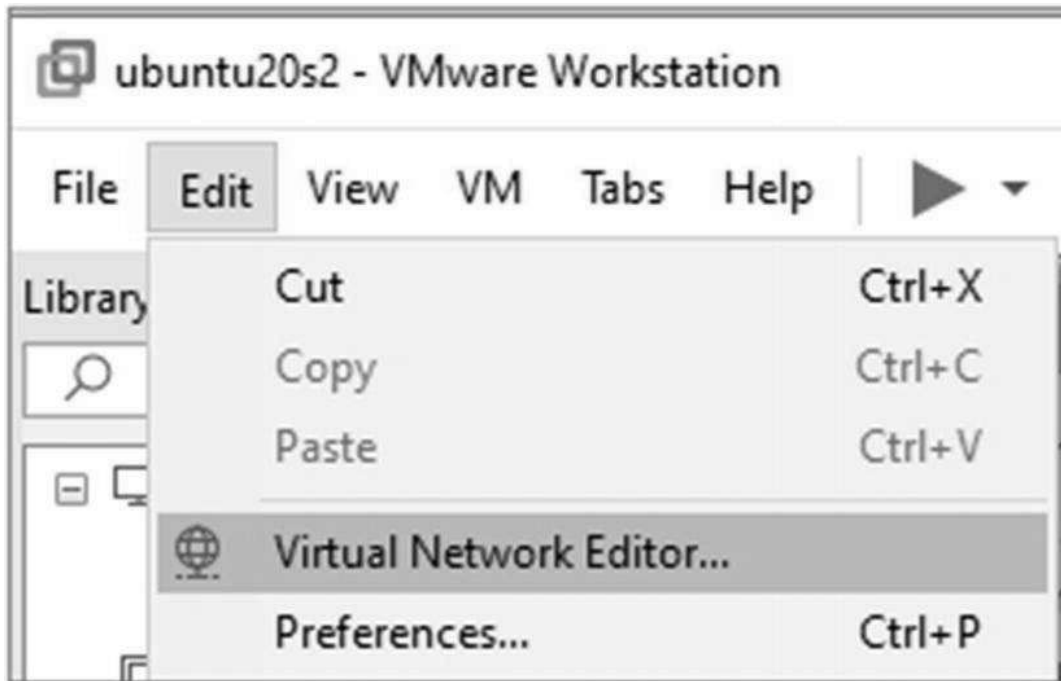


Figure 4-9. Selecting Virtual Network Editor

3 When the Virtual Network Editor opens for the first time, as expected, there are three virtual switches (VMnets). Like a real network adapter, a virtual adapter can be referred to as a virtual NIC or a virtual network interface. The Virtual Network Editor is the default control tower to manage different virtual networks, such as adding/removing networks, enabling/disable DHCP, and more (Figure [4-10](#)).

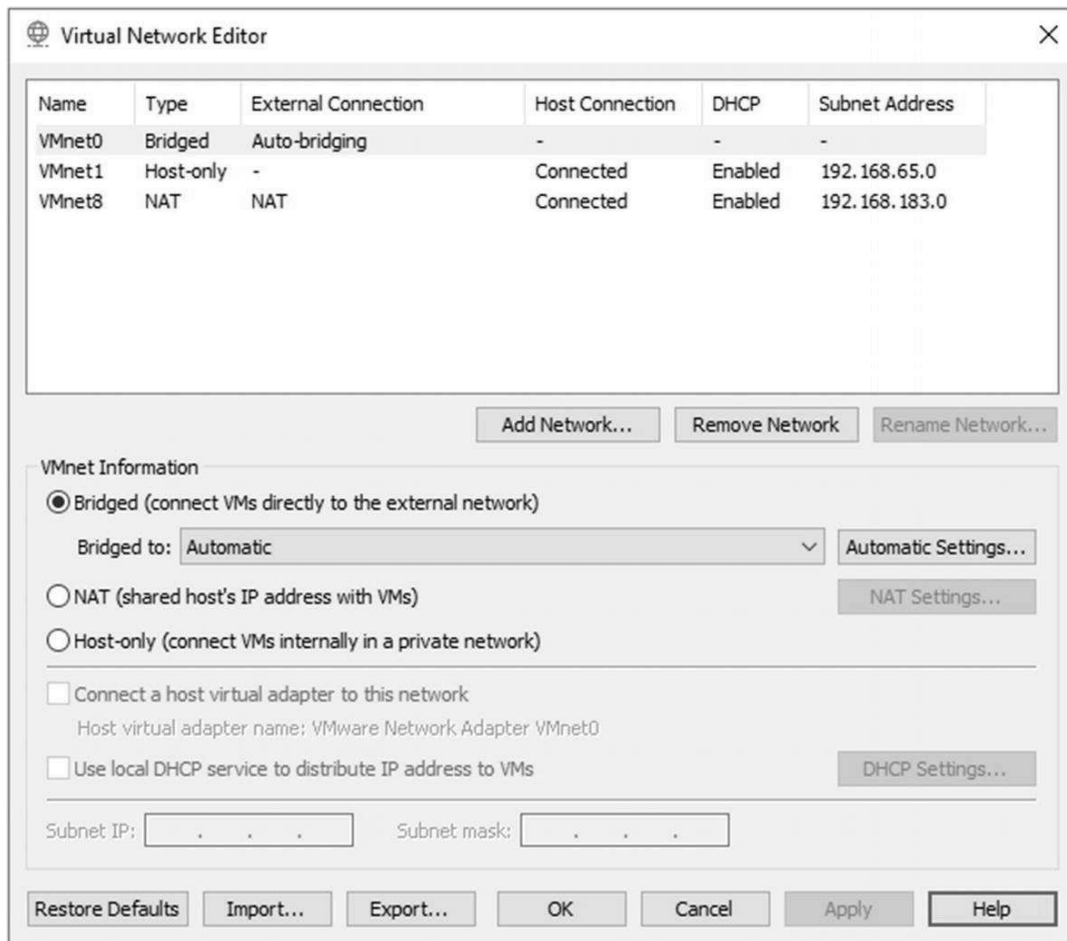


Figure 4-10. Virtual Network Editor, default adapter types



In the Virtual Network Editor window shown in Figure 4-10, click the Help button, which will open the documentation on the official VMware site to “Using the Virtual Network Editor.”

Virtual Network Interface Description

Here is a quick overview of the different network modes and networking jargon we have been discussing. If you have studied or studied Cisco CCNA, you will be already familiar with these networking terminologies in the network industry. If you are studying for CCNP or completed CCNP studies, you probably will be able to relate these network adapter modes to the real production environment. However, even if you are not familiar with the networking technologies, you will not have any trouble understanding the following explanations. As mentioned earlier, there are three out-of-the-box network adapters, but you can add or remove more adapters as required. Also, the subnets

can be specified to your preference as you build your lab. Table 3-4 describes the three virtual network modes and one user (custom) mode.

Virtual Network Adapter Modes Explained

By default, in bridged network adapter mode, all virtual machines share the host machine's network connection. By default, the bridged network uses VMnet0 in VMware Workstation, and all computers use the same gateway and hence the same DHCP and DNS server used by the host PC. Both the host PC and virtual machines acquire the same subnet's IP addresses with the same default gateway address. Figure 4-11 depicts a typical bridged network on VMware Workstation Pro.

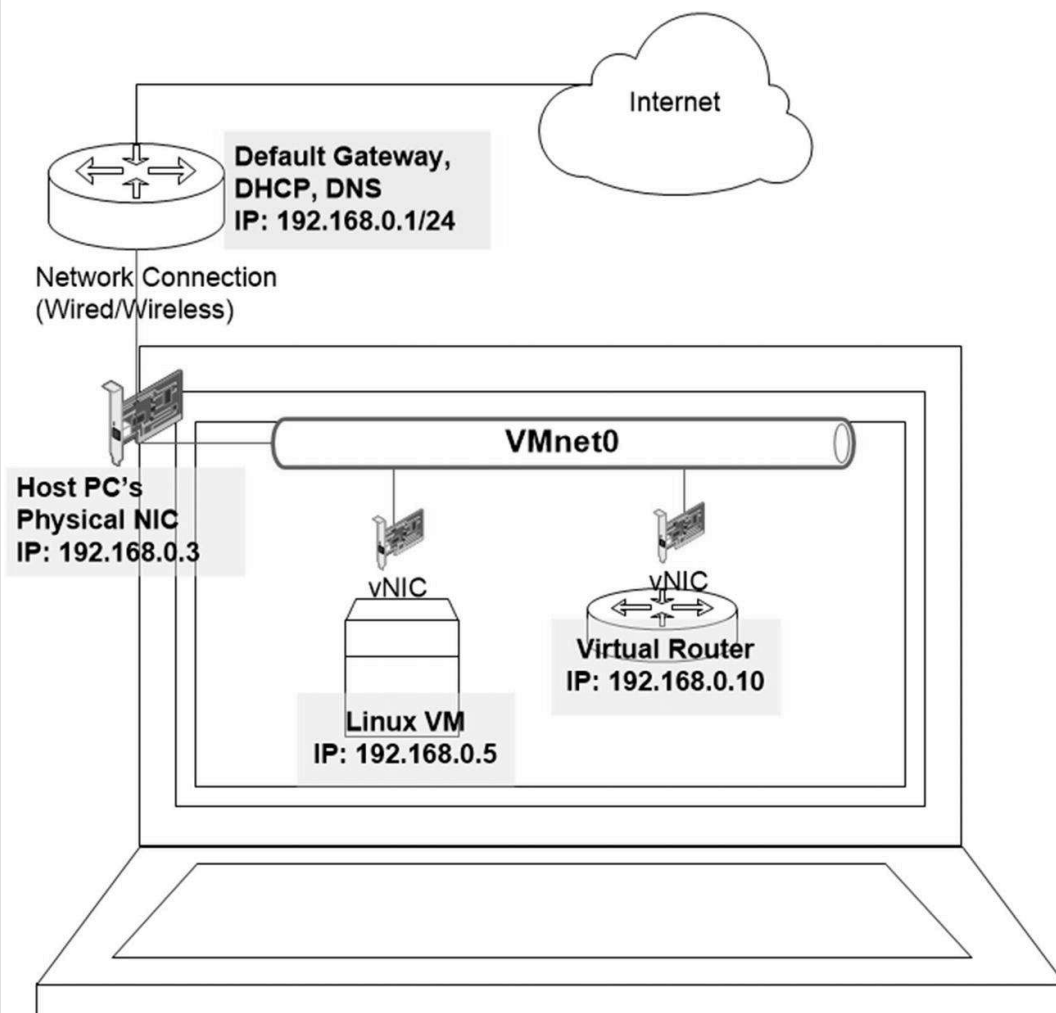


Figure 4-11. VMware network adapters, bridged mode

Host-only mode is perfect for testing two or more VMs in an isolated network within the virtual environment. By default, a host-only mode connects to VMnet1, and VMs are connected in the same subnet for experimenting with no interaction outside of the experimental network. Your

host PC and VMs can still communicate via VMnet1. In Figure [4-12](#), notice that the workstation provides the DHCP service.

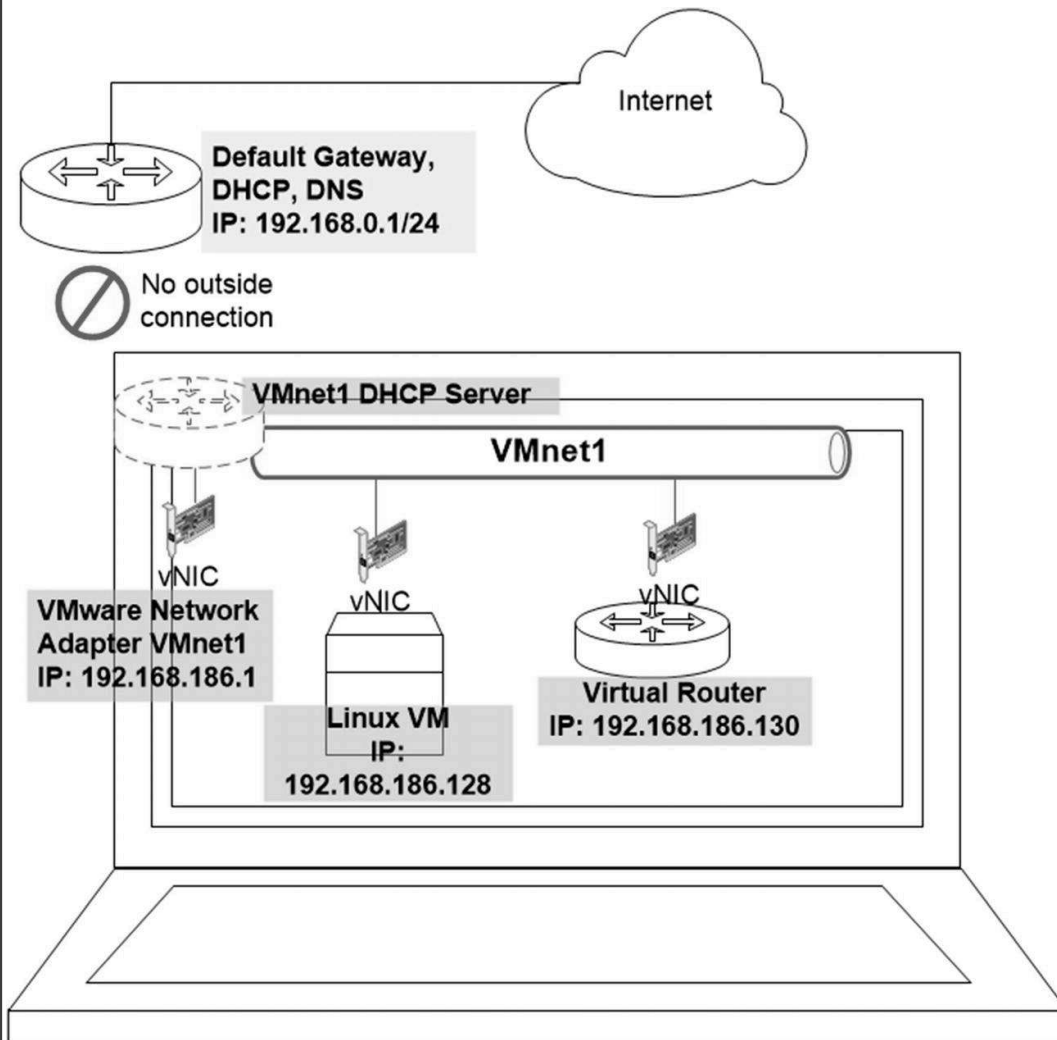


Figure 4-12. VMware network adapters, host-only mode

Network Address Translation (NAT) mode connects via VMnet8 by default. Network address translation means that the VM's internal IP address and the host's external IP address are different. In an external network, you can only see the host's subnet address. If the host can connect to the Internet, it can communicate with the Internet. Figure [4-13](#) depicts an example of a NAT configuration on VMware Workstation Pro.

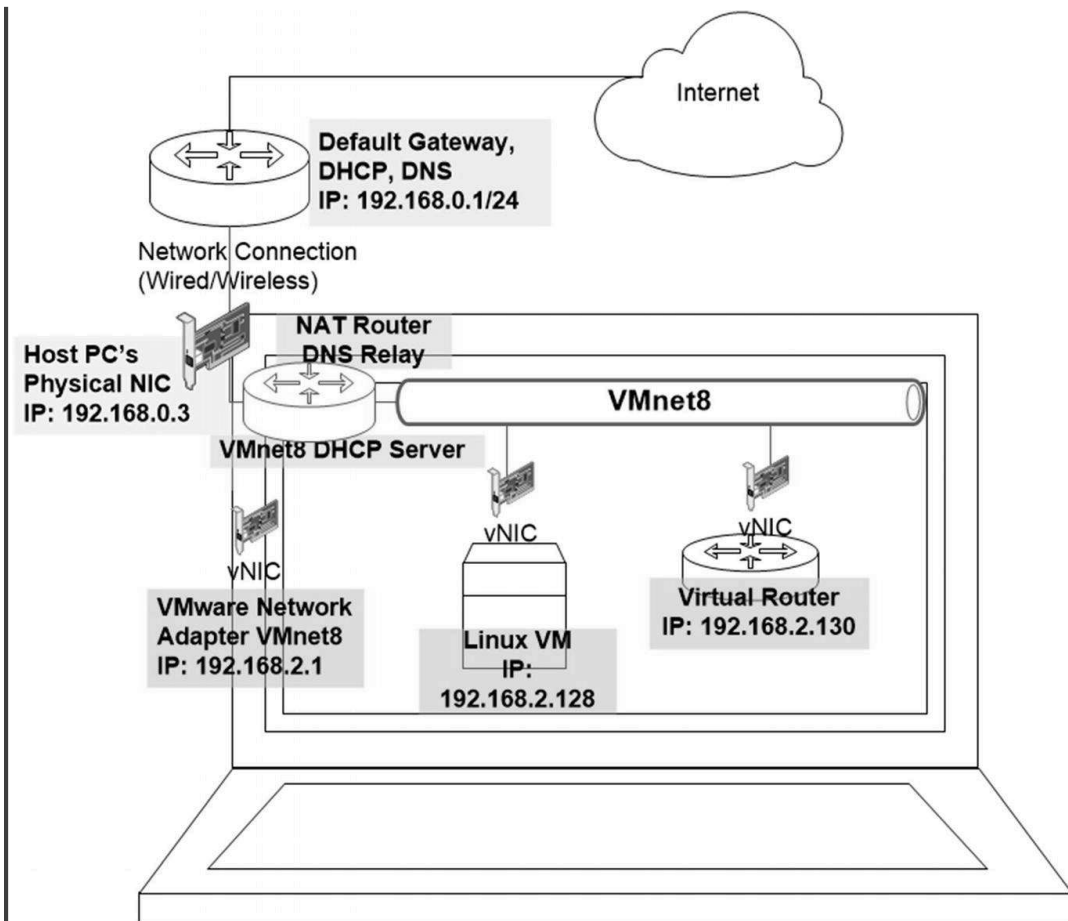


Figure 4-13. VMware network adapters, NAT mode

In custom networking, you can select one of the previous modes and customize it to suit your scenarios. It is the same mode as NAT mode but with some changes in the default settings.

You are encouraged to spend enough time understanding different virtual network modes, compared to how they are different from the real physical network, and then visualize how you can use different modes in your PoC labs. Getting familiar with how each network mode interacts with your host machine and other networks will help you build a lab that can test more advanced IT theories. Open the Virtual Network Editor from the menu and spend some time exploring the settings, adding/removing new VMnets, and customizing settings. In the following section, you will review VM settings in more detail.



Why perform a PoC? In the field of IT, the abbreviation PoC stands for *proof of concept*, and it means proving an existing or new IT concept (or a problem) in a given scenario. A PoC experimental lab

tests that a certain theory works the same way as the technology was designed for. To better understand and go beyond your colleagues, you must spend many hours in PoC labs because you will gain broader knowledge than a book or real-life scenarios can teach you. You always learn more by studying a theory and testing the theory in a PoC lab as you will not encounter all the theories or problems in the real production environment.

Revealing Each Virtual Network Interface Types

Let's look at each type.

Host-Only Networking

In this mode, your VMs share a private network with the host and other virtual devices in the same subnet. This mode is useful if your VMs do not need communications outside of your host. If you are using the default host-only networking, all VMs will connect to VMnet1, and you can think of VMnet1 as a single virtual network switch. In this mode, a built-in DHCP server provides IP addresses to hosts connecting to this network for convenience. Figures [4-14](#) to [4-18](#) display how VMnet1 is configured, and related configuration settings can be found on your host's operating system. Refer to the following tutorial for even more clarification on VMware networking concepts:

<https://rednectar.net/2011/07/20/vmware-interfaces-tutorial/>.

Figure [4-14](#) displays that the out-of-the-box VMnet1 subnet assigned to this network is 192.168.65.0/24. Of course, the subnet and CIDR can be changed as required. Also, there is an option to disable the DHCP service so you can manually assign IP addresses within a VM's OS.

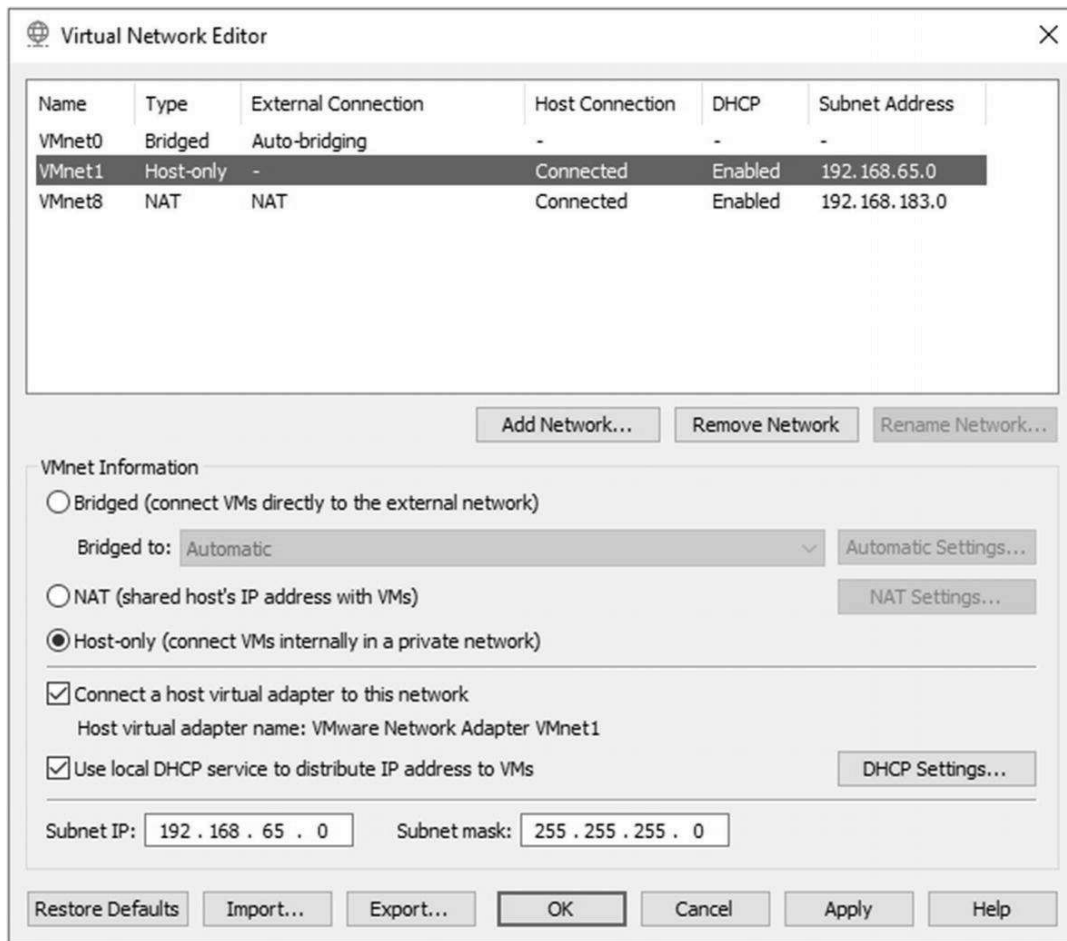


Figure 4-14. Virtual Network Editor, VMnet1 host-only network

Figure [4-15](#) shows you the VM settings where the relevant network connection is configured.

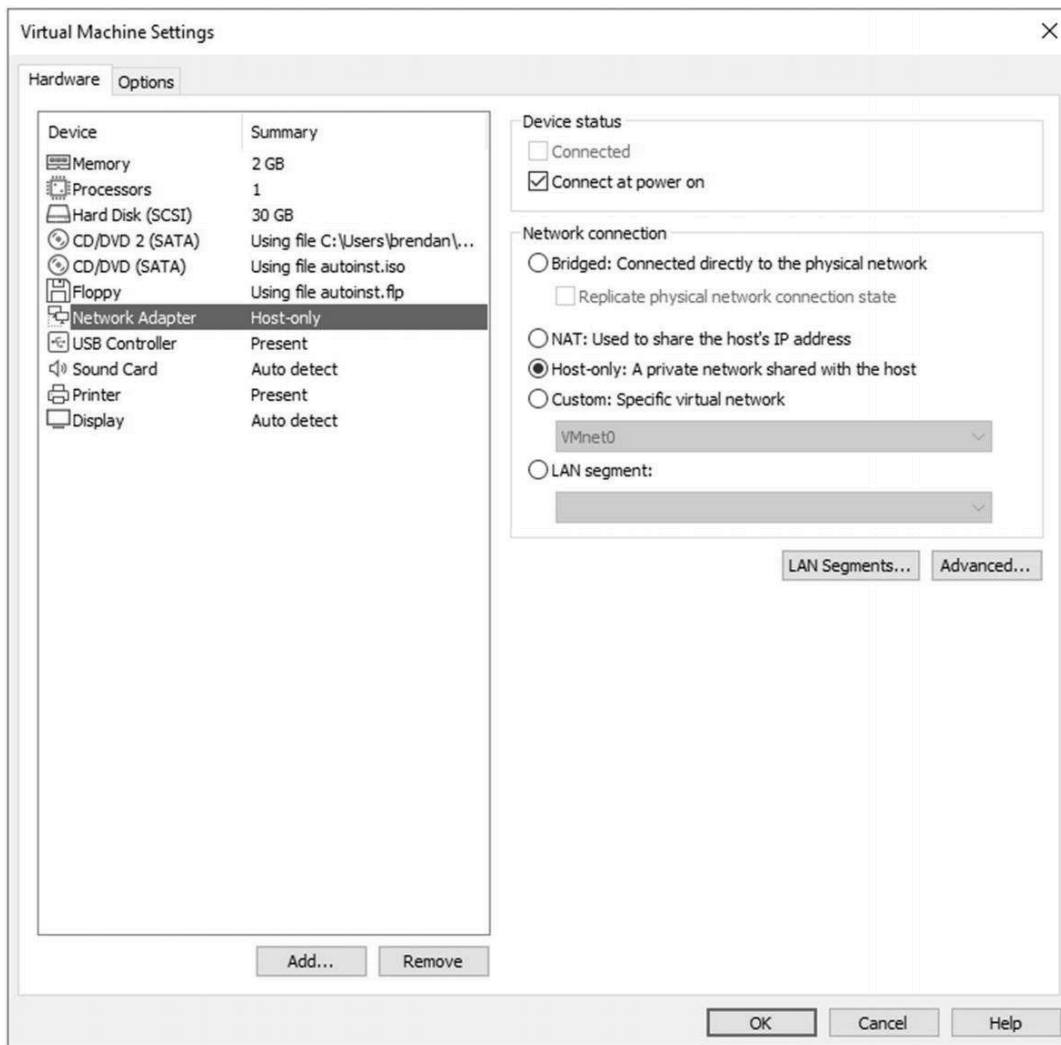


Figure 4-15. Virtual machine settings, host-only network connection

Figure 4-16 displays the result from the `netsh interface ip show addresses "VMware Network Adapter VMnet1"` command, which reveals configuration settings for VMnet1 on your host PC. As expected, the IP address assigned to the host's VMnet1 adapter is 192.168.65.1/24 with an interface metric of 35. The interface metric 35 means that the DHCP service configured this IP address.

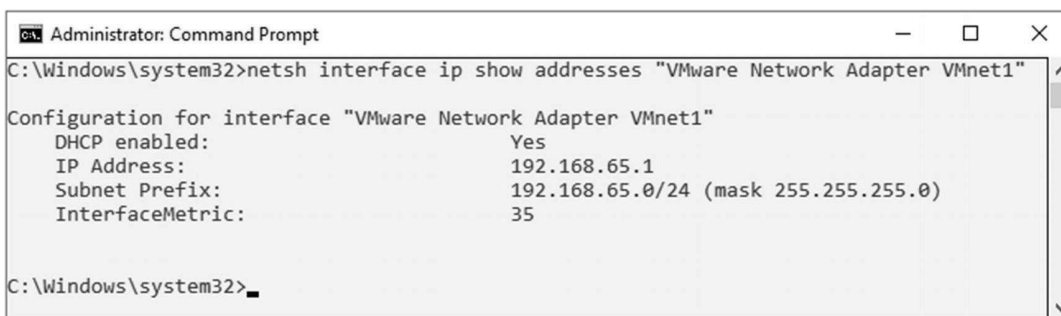
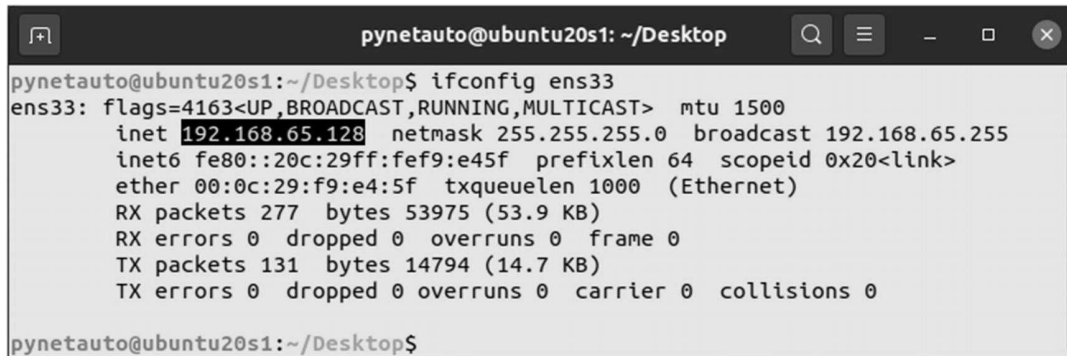


Figure 4-16. Windows host PC, VMware network adapter VMnet1

As shown in Figure 4-17, when you start a VM and check the network interface settings, the DHCP server–assigned IP to your VM will be from the same subnet as your VMnet1 network. Since you have not created and installed any VMs, you can check this setting after completing the Linux installation in Chapters 5 and 6. Note that the interface name assigned to the Ubuntu VM is ens33; the network adapter name for Linux machines depends on your Linux OS version and distro type.

A terminal window titled 'pynetauto@ubuntu20s1: ~/Desktop' showing the output of the 'ifconfig ens33' command. The output displays network details for the ens33 interface, including flags, MTU, IP address (192.168.65.128), netmask, broadcast address, MAC address, and statistics for RX and TX packets, errors, and drops.

```
pynetauto@ubuntu20s1:~/Desktop$ ifconfig ens33
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.65.128 netmask 255.255.255.0 broadcast 192.168.65.255
    inet6 fe80::20c:29ff:fe45f prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:f9:e4:5f txqueuelen 1000 (Ethernet)
    RX packets 277 bytes 53975 (53.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 131 bytes 14794 (14.7 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

pynetauto@ubuntu20s1:~/Desktop$
```

Figure 4-17. Virtual machine, ens33 network adapter

Refer to the earlier Figure 4-12 to visualize and help you understand the host-only network mode. Now you know where to find the information on the host-only virtual network for your VMs. Next, let's look at the bridged mode.

Bridged Networking

In bridged mode, a VM will act as if it is another computer on the same network as the host computer, so the VM receives an IP address from the real network. For default bridged mode, the VMnet0 network is assigned. If you are connecting to the Internet using your home network, your home network router usually provides DHCP services. By default, ISP-provided internet modems/routers will have an IP subnet of either 192.168.0.0/24 or 192.168.1.0/24, with the default gateway taking the first IP (.1) and last IP (.254) addresses on the network. In bridged mode, other physical computers can communicate to your VMs, and you can also emulate multiple computers from one computer. In bridged network mode, by default, your VMs can communicate to the Internet physically and VMs on your network.

Figure 4-18 displays the default bridged network in the Virtual Network Editor. The IP address will be assigned from the real network device on the host's physical adapter network, so the DHCP options are grayed out under the settings.

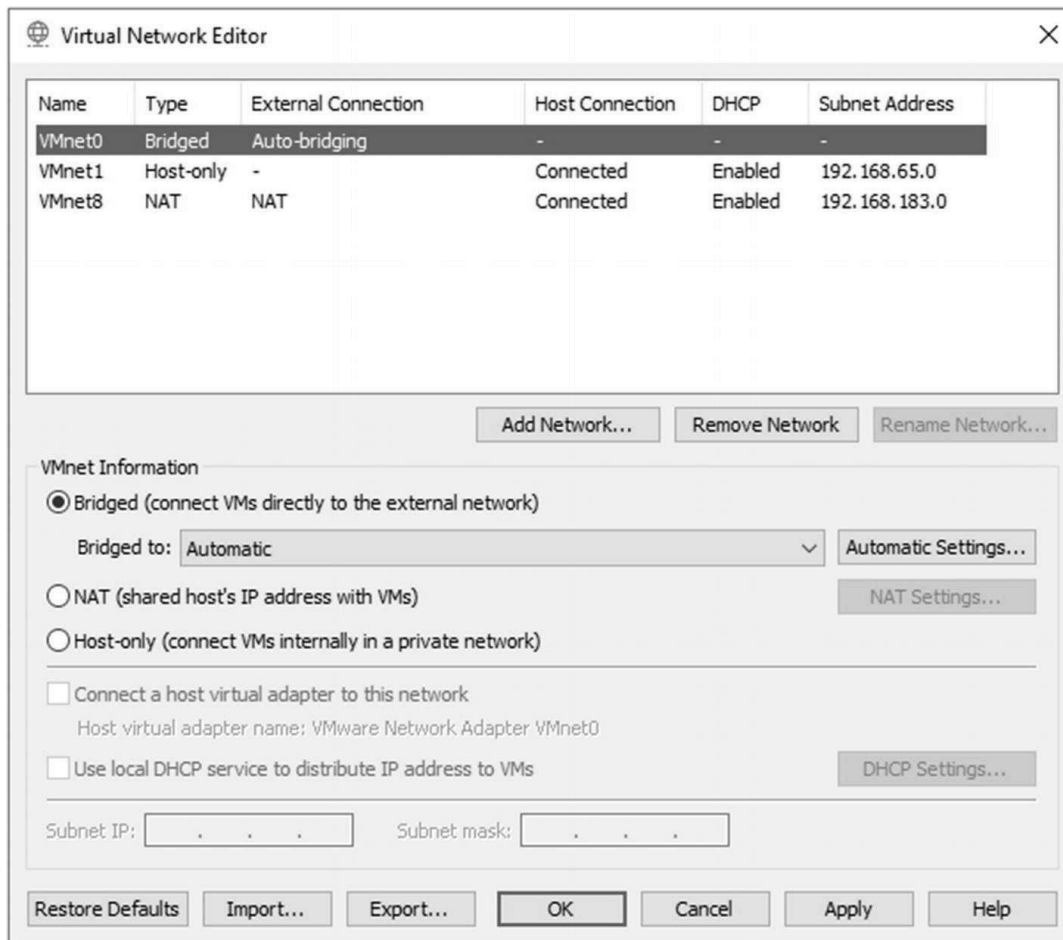


Figure 4-18. Virtual Network Editor, VMnet0 bridged network

Figure 4-19 shows the VM settings where the bridged network is configured and used.

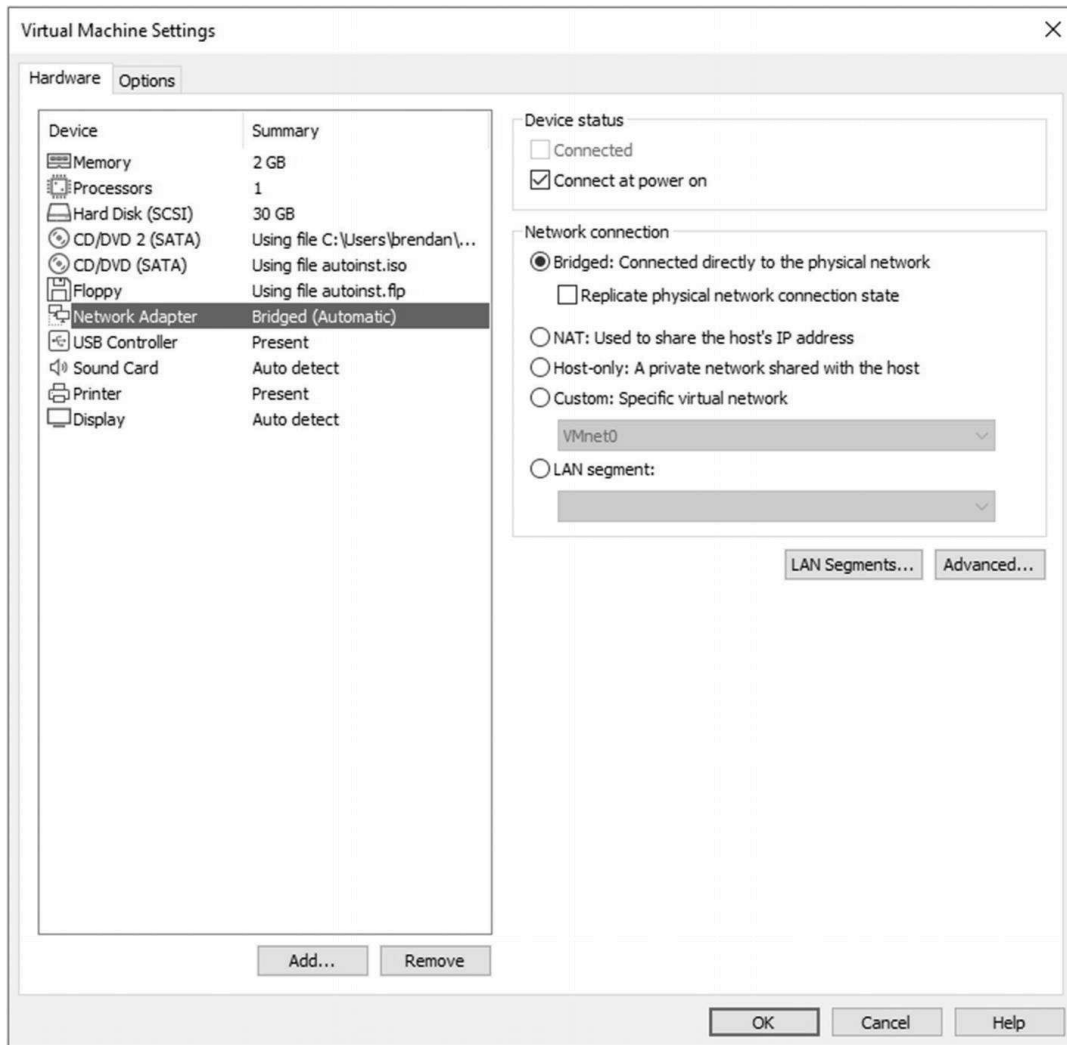


Figure 4-19. Virtual machine settings, bridged network connection

Figure [4-20](#) reveals the host's physical adapter network configuration from the DHCP server at 192.168.0.1; the default gateway doubles as a DHCP server in a typical home network. My host connects to the Internet via wireless, so a Wi-Fi 3 interface is shown as the adapter name, but your settings will differ from Figure [4-20](#).

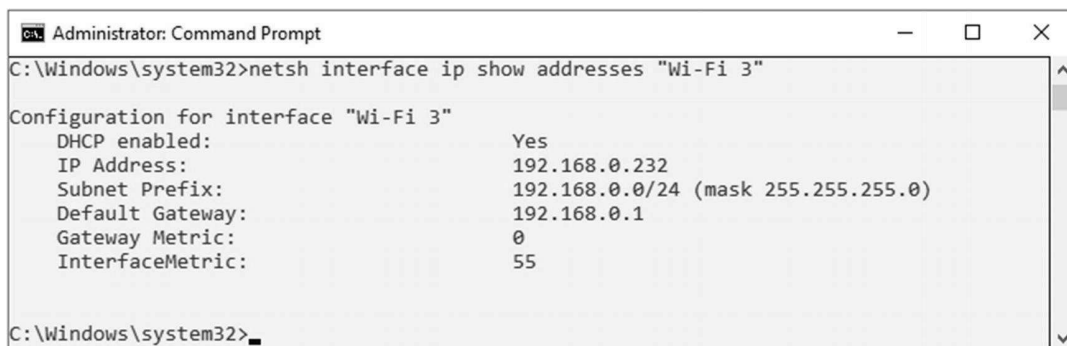
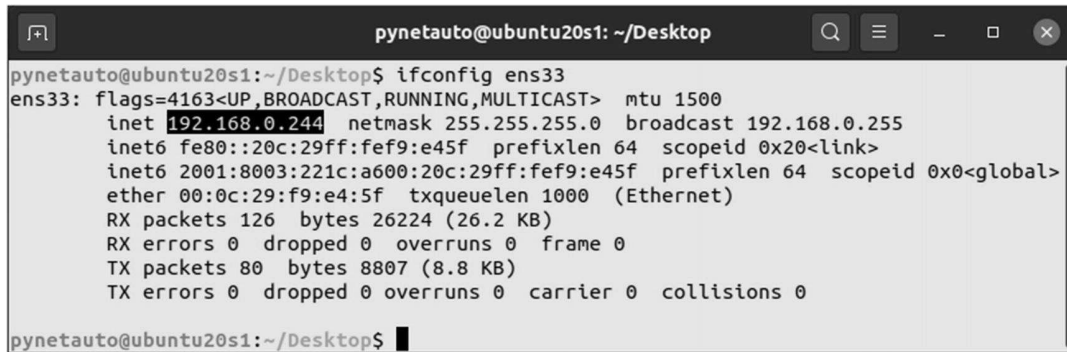


Figure 4-20. Windows host PC, local PC's internet adapter (Wi-Fi)

Figure [4-21](#) shows a VM terminal session revealing the IP address received from the home network's DHCP server. Also, refer to the previous Figure [4-11](#) to visualize and help you understand the bridged network mode.



```
pynetauto@ubuntu20s1:~/Desktop$ ifconfig ens33
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.244 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 fe80::20c:29ff:fe45f:e45f prefixlen 64 scopeid 0x20<link>
    inet6 2001:8003:221c:a600:20c:29ff:fe45f:e45f prefixlen 64 scopeid 0x0<global>
    ether 00:0c:29:f9:e4:5f txqueuelen 1000 (Ethernet)
    RX packets 126 bytes 26224 (26.2 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 80 bytes 8807 (8.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

pynetauto@ubuntu20s1:~/Desktop$
```

Figure 4-21. Virtual machine, ens33 network adapter

Network Address Translation Networking

In Network Address Translation (NAT) mode, a VM shares the host's IP and MAC addresses. The outside network (Internet) sees your network as a single network identity, and the internal network is not visible outside the network. Typically and in the traditional sense, NAT is used to hide the internal network from outsiders and extend the number of useful IPv4 IP addresses within your network. If you have already undertaken Cisco CCNA routing and switching studies, understanding NAT or PAT is part of the study curriculum, so understanding this type of networking mode will also make immediate sense. Throughout this book, you will utilize NAT's features for your entire lab configuration to keep the networking configuration to a minimum.

Let's look at Figure [4-22](#) and review how NAT is configured under VMware Workstation Pro configuration. In NAT mode, VMware Workstation provides the DHCP service, meaning that another subnet is used inside the network; in this case, the subnet assigned to NAT is in the 192.168.183.0/24 range, and this is the primary method in which our lab will connect to the Internet and host.

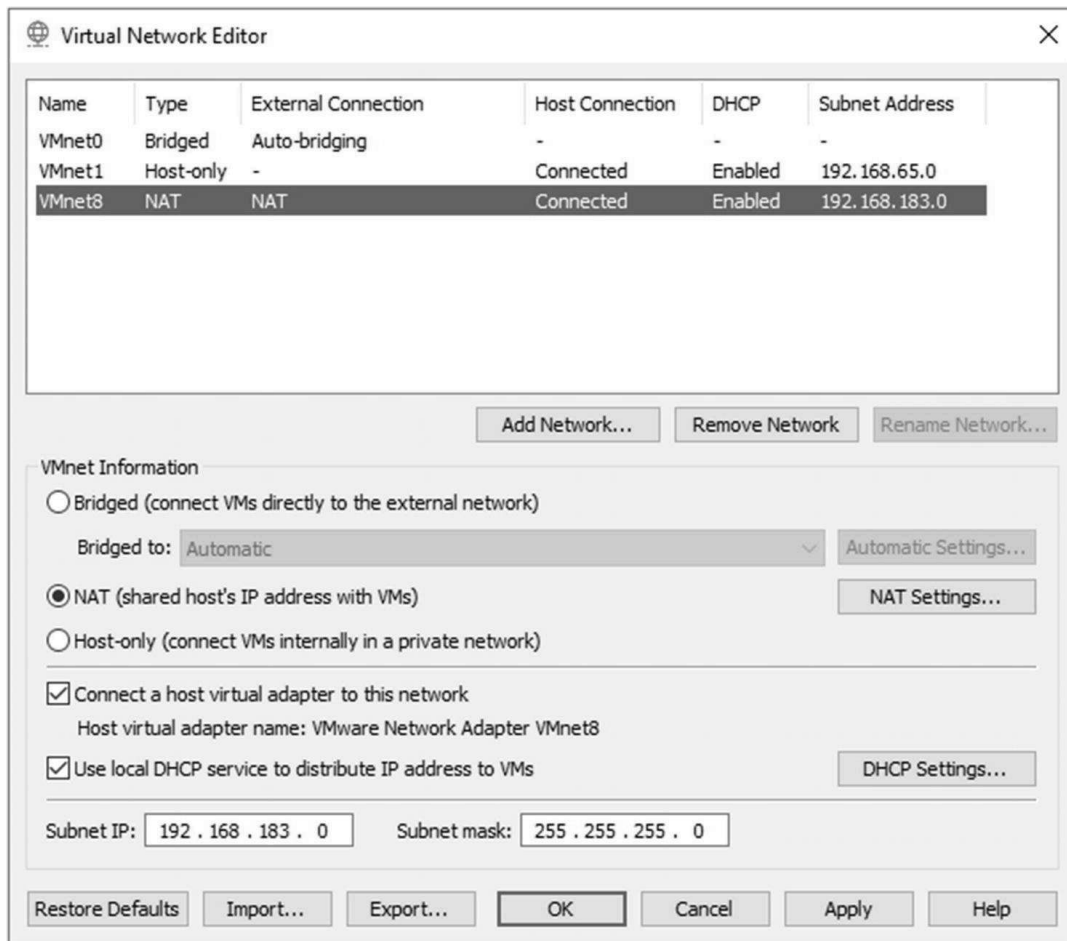


Figure 4-22. Virtual Network Editor, VMnet8 NAT adapter

Click the NAT settings button in Figure [4-21](#); it reveals NAT or VMnet8 settings below. You will notice that the gateway IP address for this subnet is 192.168.183.2, and we can assume that 192.168.183.1 would have been assigned to the VMnet8 adapter on the host PC; this will be verified in Figure [4-23](#).

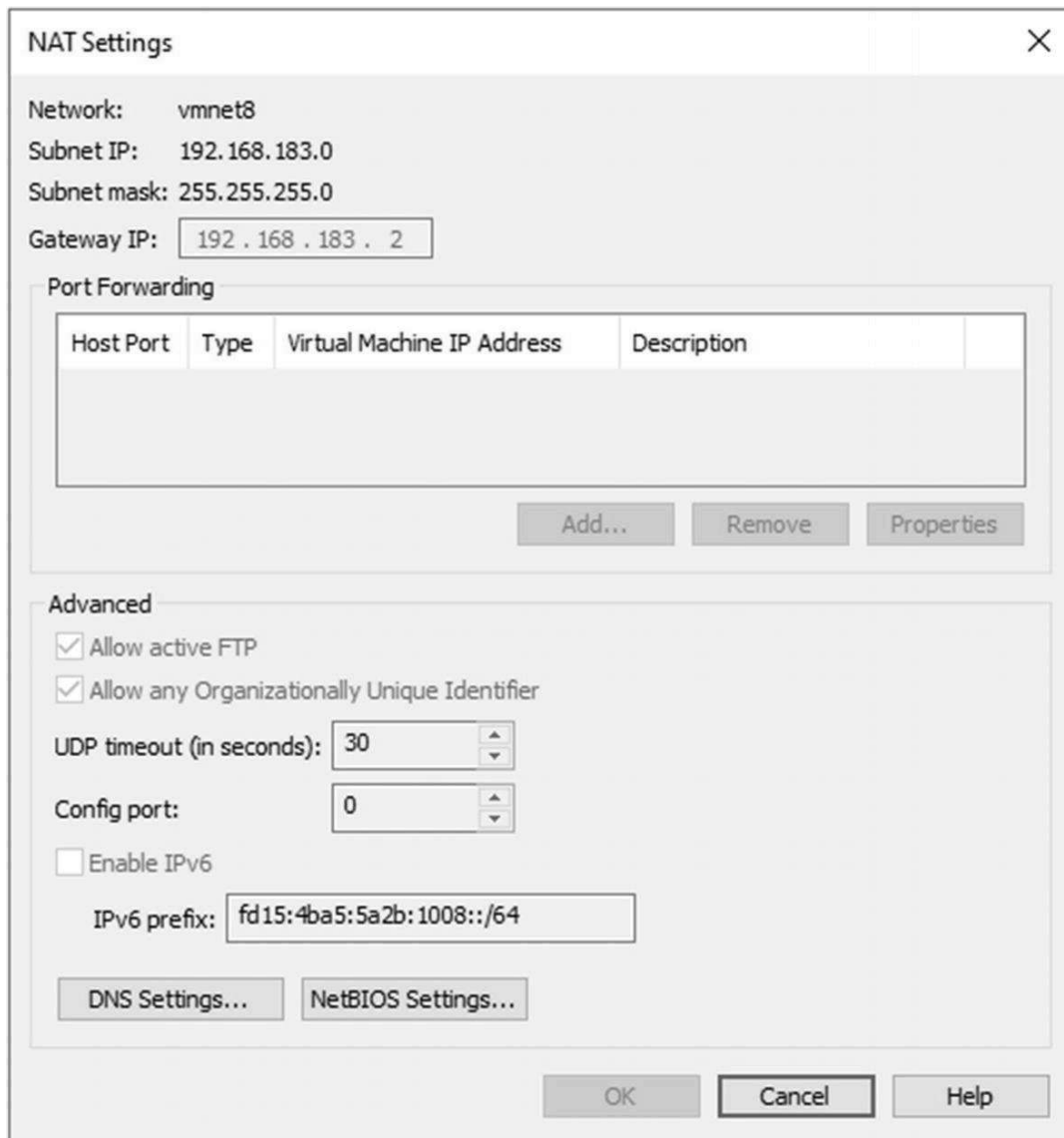


Figure 4-23. Virtual Network Editor, NAT settings

Figure [4-24](#) shows the network connection configuration of a VM.

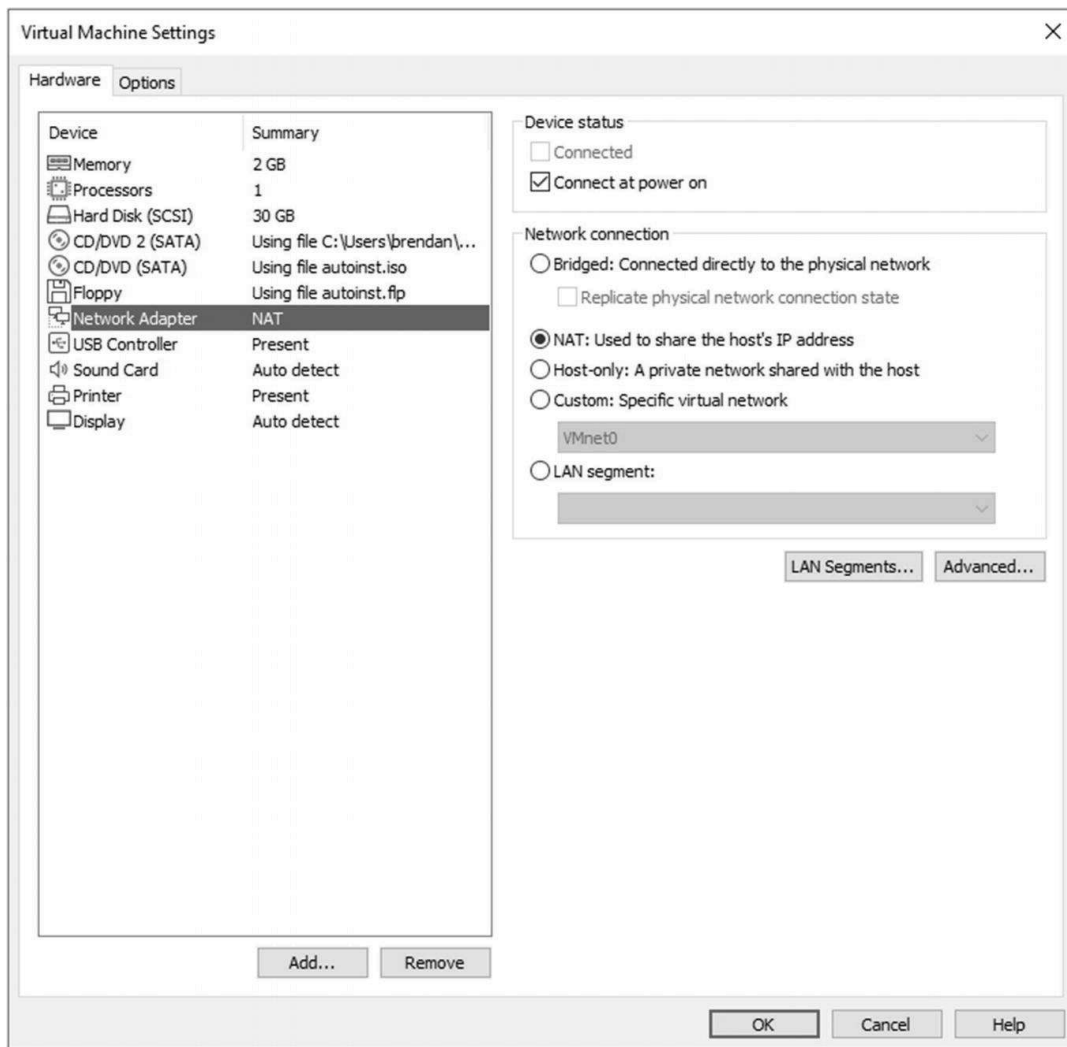


Figure 4-24. Virtual machine settings, NAT network connection

Figure 4-25 shows the IP address configuration from the DHCP server; as discussed, 192.168.183.1/24 is assigned to VMnet8 on the host. However, the default gateway for the VMs is 192.168.183.2.

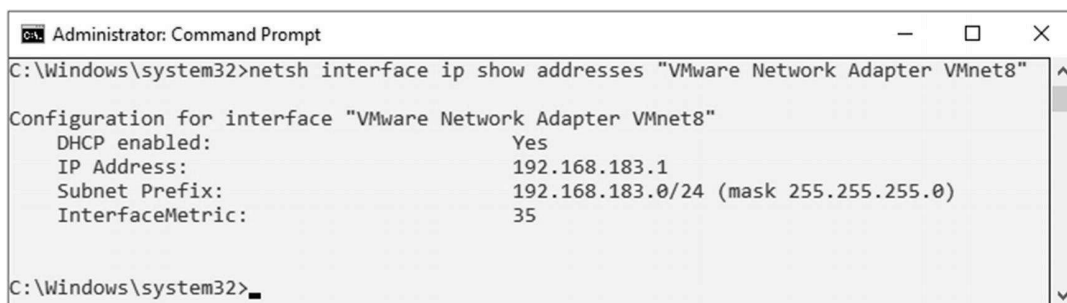
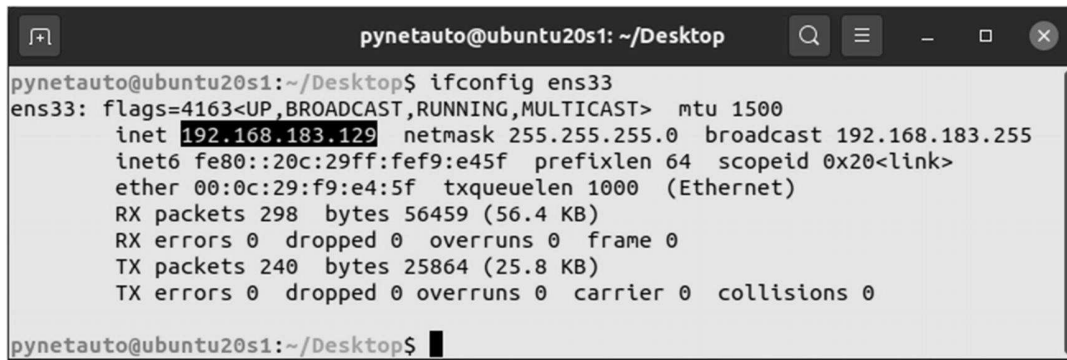


Figure 4-25. Windows host PC, VMware network adapter VMnet8

As expected, a VM on the NAT network within VMware Workstation Pro will assign an IP address from the same 192.168.183.0/24 network. In Figure 4-26, 192.168.183.129/24 has been assigned to the Linux server's ens33

network interface. Refer to the previous Figure [4-13](#) to help you better understand the NAT mode.

A terminal window titled 'pynetauto@ubuntu20s1: ~/Desktop' showing the output of the 'ifconfig ens33' command. The output displays the configuration for the ens33 network interface, including its flags, MTU, IP address (192.168.183.129), netmask (255.255.255.0), broadcast address (192.168.183.255), and various statistics for RX and TX packets, bytes, errors, and drops. The interface is identified as an Ethernet card with a txqueuelen of 1000.

```
pynetauto@ubuntu20s1:~/Desktop$ ifconfig ens33
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.183.129  netmask 255.255.255.0  broadcast 192.168.183.255
    inet6 fe80::20c:29ff:fe45f  prefixlen 64  scopeid 0x20<link>
    ether 00:0c:29:f9:e4:5f  txqueuelen 1000  (Ethernet)
    RX packets 298  bytes 56459 (56.4 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 240  bytes 25864 (25.8 KB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

pynetauto@ubuntu20s1:~/Desktop$
```

Figure 4-26. Virtual machine, ens33 network adapter

Summary

Virtualization technology makes cloud services possible, so we discussed the difference between Type-1 and Type-2 hypervisors of current virtualization technology. This chapter also introduced you to VMware Workstation 15 Pro and why we are using this solution over others. We also looked at this software's basic operations on our Windows PC and delved into different networking modes that users can use to build a virtual lab environment for our work and leisure use. Now that you know the VMware Workstation essential features, it is time to download, install, and create virtual machines, namely, Ubuntu 20.04 LTS (Chapter [5](#)) and CentOS 8.1 servers (Chapter [6](#)).

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